

=== Training Inclination Model ===

Inclination - Starting hyperparameter tuning...

```

=====
=====|
| Iter | Eval   | Objective   | Objective   | BestSoFar   | BestSoFar   |
NumLearningC-| MinLeafSize | MaxNumSplits |
|       | result  |             | runtime     | (observed)  | (estim.)    | cycles
|       |         |             |             |             |             |
=====
=====|
| 1 | Best | 0.16667 | 0.64609 | 0.16667 | 0.16667 |
27 | 6 | 51 |
| 2 | Accept | 0.31597 | 2.2565 | 0.16667 | 0.1775 |
173 | 1 | 3 |
| 3 | Accept | 0.47222 | 2.9658 | 0.16667 | 0.18229 |
241 | 49 | 55 |
| 4 | Accept | 0.48611 | 0.21276 | 0.16667 | 0.1667 |
15 | 89 | 4 |
| 5 | Accept | 0.49653 | 5.7538 | 0.16667 | 0.18387 |
499 | 4 | 1 |
| 6 | Best | 0.15625 | 0.34647 | 0.15625 | 0.15802 |
26 | 6 | 198 |
| 7 | Accept | 0.17014 | 0.25473 | 0.15625 | 0.16458 |
18 | 6 | 285 |
| 8 | Accept | 0.27778 | 0.24545 | 0.15625 | 0.16182 |
18 | 25 | 26 |
| 9 | Best | 0.15278 | 0.36815 | 0.15278 | 0.15307 |
26 | 2 | 85 |
| 10 | Accept | 0.16667 | 0.1617 | 0.15278 | 0.15166 |
10 | 1 | 285 |
| 11 | Accept | 0.28819 | 0.15401 | 0.15278 | 0.15269 |
10 | 1 | 5 |
| 12 | Accept | 0.17708 | 0.61843 | 0.15278 | 0.15271 |
45 | 1 | 287 |
| 13 | Accept | 0.15625 | 0.31418 | 0.15278 | 0.15275 |
23 | 2 | 260 |
| 14 | Accept | 0.17014 | 0.50071 | 0.15278 | 0.15325 |
38 | 3 | 124 |
| 15 | Best | 0.15278 | 0.25992 | 0.15278 | 0.15261 |
18 | 3 | 121 |
| 16 | Accept | 0.15625 | 0.27414 | 0.15278 | 0.1527 |
20 | 1 | 140 |
| 17 | Accept | 0.16667 | 0.29599 | 0.15278 | 0.15496 |
22 | 3 | 111 |
| 18 | Accept | 0.15972 | 0.27545 | 0.15278 | 0.15571 |
20 | 2 | 196 |
| 19 | Accept | 0.16667 | 5.7606 | 0.15278 | 0.15339 |
494 | 1 | 281 |

```

```

| 20 | Accept | 0.15278 | 0.16135 | 0.15278 | 0.15448 | ✓
10 | 5 | 100 |
|=====|
| Iter | Eval | Objective | Objective | BestSoFar | BestSoFar | ✓
NumLearningC-| MinLeafSize | MaxNumSplits |
| result | runtime | (observed) | (estim.) | cycles ✓
| | |
|=====|
| 21 | Accept | 0.18403 | 0.16044 | 0.15278 | 0.15637 | ✓
10 | 5 | 124 |
| 22 | Accept | 0.16667 | 1.5658 | 0.15278 | 0.15657 | ✓
131 | 1 | 100 |
| 23 | Accept | 0.69097 | 0.15656 | 0.15278 | 0.15568 | ✓
10 | 136 | 279 |
| 24 | Accept | 0.15972 | 4.8871 | 0.15278 | 0.15579 | ✓
427 | 9 | 94 |
| 25 | Accept | 0.15278 | 1.8892 | 0.15278 | 0.15366 | ✓
160 | 2 | 126 |
| 26 | Accept | 0.16667 | 0.54738 | 0.15278 | 0.15658 | ✓
43 | 2 | 126 |
| 27 | Accept | 0.15972 | 3.8847 | 0.15278 | 0.15657 | ✓
337 | 8 | 88 |
| 28 | Accept | 0.49653 | 4.3131 | 0.15278 | 0.15686 | ✓
401 | 27 | 1 |
| 29 | Accept | 0.69097 | 3.9492 | 0.15278 | 0.15715 | ✓
388 | 144 | 1 |
| 30 | Accept | 0.32292 | 4.8858 | 0.15278 | 0.15722 | ✓
439 | 30 | 287 |

```

Optimization completed.

MaxObjectiveEvaluations of 30 reached.

Total function evaluations: 30

Total elapsed time: 52.836 seconds

Total objective function evaluation time: 48.0656

Best observed feasible point:

NumLearningCycles	MinLeafSize	MaxNumSplits
18	3	121

Observed objective function value = 0.15278

Estimated objective function value = 0.16317

Function evaluation time = 0.25992

Best estimated feasible point (according to models):

NumLearningCycles	MinLeafSize	MaxNumSplits
-------------------	-------------	--------------

43

2

126

Estimated objective function value = 0.15722

Estimated function evaluation time = 0.58592

Inclination - Tuned 10-Fold Cross-Validated Accuracy: 86.11%

Inclination - Tuned 10-Fold Macro F1-Score: 0.83

Inclination - Test Accuracy: 88.73%

Inclination - Test Macro F1-Score: 0.84

=== Training Frequency Model ===

Frequency - Starting hyperparameter tuning...

```

=====
| Iter | Eval | Objective | Objective | BestSoFar | BestSoFar |
NumLearningC-| MinLeafSize | MaxNumSplits |
| result | runtime | (observed) | (estim.) | cycles
=====
| 1 | Best | 0.40993 | 0.86105 | 0.40993 | 0.40993 |
50 | 25 | 172 |
| 2 | Accept | 0.78384 | 1.262 | 0.40993 | 0.43049 |
94 | 1 | 1 |
| 3 | Accept | 0.52093 | 0.18777 | 0.40993 | 0.42384 |
10 | 1 | 10 |
| 4 | Best | 0.23953 | 7.5116 | 0.23953 | 0.30016 |
460 | 4 | 780 |
| 5 | Accept | 0.42259 | 6.2518 | 0.23953 | 0.23957 |
425 | 28 | 1025 |
| 6 | Best | 0.22298 | 1.8978 | 0.22298 | 0.22296 |
113 | 1 | 1021 |
| 7 | Best | 0.22298 | 1.192 | 0.22298 | 0.22291 |
70 | 1 | 327 |
| 8 | Accept | 0.22785 | 4.0294 | 0.22298 | 0.223 |
244 | 2 | 505 |
| 9 | Accept | 0.24051 | 1.1184 | 0.22298 | 0.22594 |
63 | 1 | 644 |
| 10 | Accept | 0.23174 | 1.5946 | 0.22298 | 0.22652 |
95 | 2 | 622 |
| 11 | Accept | 0.2298 | 2.663 | 0.22298 | 0.22378 |
159 | 1 | 1004 |
| 12 | Accept | 0.2259 | 7.3009 | 0.22298 | 0.22457 |
446 | 1 | 132 |
| 13 | Accept | 0.24051 | 0.83694 | 0.22298 | 0.22614 |
47 | 2 | 201 |

```

```

| 14 | Accept | 0.22785 | 8.1017 | 0.22298 | 0.22552 | ✓
472 | 1 | 284 |
| 15 | Accept | 0.22785 | 0.70065 | 0.22298 | 0.22552 | ✓
38 | 2 | 1024 |
| 16 | Accept | 0.24051 | 0.66877 | 0.22298 | 0.22412 | ✓
36 | 1 | 196 |
| 17 | Accept | 0.22882 | 8.2716 | 0.22298 | 0.22512 | ✓
484 | 2 | 984 |
| 18 | Accept | 0.22687 | 8.0199 | 0.22298 | 0.22576 | ✓
484 | 1 | 260 |
| 19 | Accept | 0.22687 | 8.2443 | 0.22298 | 0.22587 | ✓
499 | 1 | 1002 |
| 20 | Accept | 0.22882 | 6.5874 | 0.22298 | 0.22596 | ✓
399 | 2 | 1025 |
|===== ✓
=====|
| Iter | Eval | Objective | Objective | BestSoFar | BestSoFar | ✓
NumLearningC-| MinLeafSize | MaxNumSplits |
| result | runtime | (observed) | (estim.) | cycles ✓
| | | | | |
|===== ✓
=====|
| 21 | Accept | 0.2298 | 8.1986 | 0.22298 | 0.22371 | ✓
494 | 1 | 251 |
| 22 | Accept | 0.28432 | 1.3783 | 0.22298 | 0.22642 | ✓
87 | 1 | 64 |
| 23 | Accept | 0.23466 | 7.638 | 0.22298 | 0.22669 | ✓
469 | 2 | 984 |
| 24 | Accept | 0.82181 | 4.8442 | 0.22298 | 0.22664 | ✓
408 | 511 | 1 |
| 25 | Accept | 0.22882 | 8.2099 | 0.22298 | 0.22686 | ✓
495 | 1 | 381 |
| 26 | Accept | 0.22882 | 7.1396 | 0.22298 | 0.22498 | ✓
429 | 1 | 1024 |
| 27 | Accept | 0.24051 | 1.2804 | 0.22298 | 0.2269 | ✓
75 | 2 | 348 |
| 28 | Accept | 0.82181 | 4.4329 | 0.22298 | 0.22685 | ✓
374 | 513 | 1015 |
| 29 | Accept | 0.82181 | 4.7475 | 0.22298 | 0.22682 | ✓
399 | 513 | 24 |
| 30 | Accept | 0.78481 | 6.3839 | 0.22298 | 0.22683 | ✓
493 | 22 | 1 |

```

Optimization completed.

MaxObjectiveEvaluations of 30 reached.

Total function evaluations: 30

Total elapsed time: 133.7405 seconds

Total objective function evaluation time: 131.5547

Best observed feasible point:

NumLearningCycles	MinLeafSize	MaxNumSplits
70	1	327

Observed objective function value = 0.22298

Estimated objective function value = 0.23158

Function evaluation time = 1.192

Best estimated feasible point (according to models):

NumLearningCycles	MinLeafSize	MaxNumSplits
484	1	260

Estimated objective function value = 0.22683

Estimated function evaluation time = 8.0391

Frequency - Tuned 10-Fold Cross-Validated Accuracy: 78.58%

Frequency - Tuned 10-Fold Macro F1-Score: 0.79

Frequency - Test Accuracy: 76.95%

Frequency - Test Macro F1-Score: 0.76

=== Training Damper Model ===

Damper - Starting hyperparameter tuning...

```

=====
| Iter | Eval | Objective | Objective | BestSoFar | BestSoFar |
NumLearningC-| MinLeafSize | MaxNumSplits |
|      | result |      runtime | (observed) | (estim.) | ycles
|      |      |      |      |      |      |
=====
|  1 | Best | 0.34247 | 0.22886 | 0.34247 | 0.34247 |
15 |      | 1 | 1 |
|  2 | Accept | 0.34589 | 1.6662 | 0.34247 | 0.34391 |
150 |      | 4 | 1 |
|  3 | Best | 0.14384 | 0.28913 | 0.14384 | 0.14389 |
21 |      | 11 | 91 |
|  4 | Accept | 0.34247 | 3.1929 | 0.14384 | 0.14391 |
296 |      | 97 | 58 |
|  5 | Accept | 0.15068 | 5.1676 | 0.14384 | 0.14456 |
435 |      | 1 | 90 |
|  6 | Accept | 0.34932 | 0.18192 | 0.14384 | 0.14386 |
13 |      | 77 | 104 |
|  7 | Accept | 0.15068 | 3.4929 | 0.14384 | 0.14387 |
304 |      | 4 | 117 |

```

```

|      8 | Accept |      0.15753 |      0.77778 |      0.14384 |      0.14387 | ✓
64 |      6 |      44 |
|      9 | Accept |      0.14726 |      1.8554 |      0.14384 |      0.14387 | ✓
156 |      1 |      291 |
|     10 | Accept |      0.15753 |      0.85327 |      0.14384 |      0.1439 | ✓
70 |      8 |      287 |
|     11 | Accept |      0.16438 |      0.23727 |      0.14384 |      0.1439 | ✓
17 |      2 |      280 |
|     12 | Accept |      0.15753 |      0.39576 |      0.14384 |      0.14823 | ✓
31 |      8 |      103 |
|     13 | Accept |      0.15411 |      4.9584 |      0.14384 |      0.14832 | ✓
417 |      1 |      173 |
|     14 | Accept |      0.15068 |      1.5331 |      0.14384 |      0.14918 | ✓
132 |      7 |      110 |
|     15 | Accept |      0.14384 |      1.8222 |      0.14384 |      0.14966 | ✓
157 |      2 |      79 |
|     16 | Accept |      0.15411 |      5.5801 |      0.14384 |      0.14945 | ✓
458 |      2 |      77 |
|     17 | Accept |      0.16096 |      0.16729 |      0.14384 |      0.14974 | ✓
10 |      1 |      148 |
|     18 | Accept |      0.14726 |      16.997 |      0.14384 |      0.14962 | ✓
487 |     10 |      84 |
|     19 | Accept |      0.15068 |      37.344 |      0.14384 |      0.14772 | ✓
482 |     11 |      52 |
|     20 | Accept |      0.15068 |      17.209 |      0.14384 |      0.14851 | ✓
384 |     10 |      88 |
|===== ✓
=====|
| Iter | Eval | Objective | Objective | BestSoFar | BestSoFar | ✓
NumLearningC-| MinLeafSize | MaxNumSplits |
|      | result |      | runtime | (observed) | (estim.) | cycles ✓
|      |      |      |      |      |      |
|===== ✓
=====|
|     21 | Accept |      0.14384 |      5.4813 |      0.14384 |      0.14844 | ✓
471 |      3 |      74 |
|     22 | Accept |      0.14726 |      5.5277 |      0.14384 |      0.14674 | ✓
476 |      3 |      66 |
|     23 | Accept |      0.57192 |      4.6539 |      0.14384 |      0.14724 | ✓
476 |     146 |      1 |
|     24 | Accept |      0.15411 |      5.5709 |      0.14384 |      0.14729 | ✓
478 |      1 |      30 |
|     25 | Accept |      0.14726 |      5.2861 |      0.14384 |      0.147 | ✓
457 |      2 |      40 |
|     26 | Accept |      0.15068 |      5.8234 |      0.14384 |      0.14748 | ✓
481 |      2 |      49 |
|     27 | Accept |      0.14726 |      5.9186 |      0.14384 |      0.14675 | ✓
496 |      3 |      58 |
|     28 | Accept |      0.15068 |      1.0394 |      0.14384 |      0.1473 | ✓

```

```

87 |           16 |           129 |
| 29 | Accept | 0.15753 | 5.6632 | 0.14384 | 0.14704 | ✓
493 |           13 |           185 |
| 30 | Accept | 0.15411 | 1.1463 | 0.14384 | 0.14717 | ✓
94 |           1 |           69 |

```

Optimization completed.

MaxObjectiveEvaluations of 30 reached.

Total function evaluations: 30

Total elapsed time: 152.3465 seconds

Total objective function evaluation time: 150.0607

Best observed feasible point:

NumLearningCycles	MinLeafSize	MaxNumSplits
21	11	91

Observed objective function value = 0.14384

Estimated objective function value = 0.14878

Function evaluation time = 0.28913

Best estimated feasible point (according to models):

NumLearningCycles	MinLeafSize	MaxNumSplits
476	3	66

Estimated objective function value = 0.14717

Estimated function evaluation time = 5.587

Damper - Tuned 10-Fold Cross-Validated Accuracy: 84.25%

Damper - Tuned 10-Fold Macro F1-Score: 0.71

Damper - Test Accuracy: 84.72%

Damper - Test Macro F1-Score: NaN

>>